

Compound solids and real measurement [Cambridge insert]

Cut a real object into named solids, add or subtract, and carry the bounds through.

LESSON 48–50

3 × 40 MIN

GEOMETRY 11, AMALIY MASHG‘ULOT

IGCSE E13.X COMPOUND SOLIDS · E13.3 BOUNDS

LESSON CLOCK

12

25'

25'

30'

30'

13'

Reading an object as solids	12 min
Adding and subtracting volume	25 min
Which faces survive	25 min
Bounds through a compound	30 min
Measuring the classroom	30 min
Homework	13 min

LEARNING OBJECTIVES

- Decompose a real object into prisms, cylinders, cones, pyramids and spheres.
- Add and subtract volumes, and decide which faces survive into the surface area.
- Carry upper and lower bounds through a compound calculation.
- Report a measured answer to a sensible degree of accuracy.

TEXTBOOK

National	pp. 241–248
Cambridge	Core & Extended, pp. 365–374
Workbook	Exercise 13.5

01

Explanation

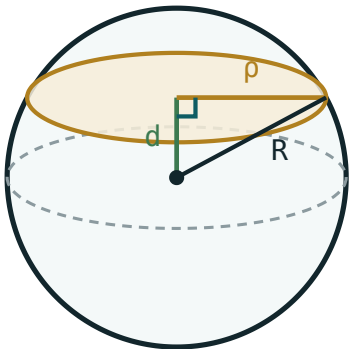
Reading an object as solids

Nothing new is needed for the next three lessons — only the seven formulas already proved. The whole skill is **reading**: seeing a grain silo as a cylinder with a hemisphere on top, a pencil as a cylinder with a cone, a water tower as a sphere on a thin cylinder.

OBJECT	READ AS	VOLUME
grain silo	cylinder + hemisphere	$\pi r^2 h + \frac{2}{3} \pi r^3$
pencil	cylinder + cone	$\pi r^2 h + \frac{1}{3} \pi r^2 k$
ice-cream	cone + hemisphere	$\frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$
a bead	sphere – cylinder	$\frac{4}{3} \pi R^3 - \pi r^2 \cdot 2R$
a pipe	cylinder – cylinder	$\pi(R^2 - r^2)L$
a mosque dome	hemisphere – hemisphere	$\frac{2}{3} \pi(R^3 - r^3)$

THE ONE RULE FOR THE CUT

Cut along a plane where two named solids meet, and the cut face is always a **circle** or a **polygon** shared by both. If your cut leaves a face that is neither, you have cut in the wrong place.



the section is a circle of radius ρ
 $\rho^2 = R^2 - d^2$

*A plane cut through a solid of revolution leaves a circle
— the shared face of the two parts.*

Adding and subtracting volume

Volume is easy: add the parts, or subtract the hollow. Nothing is shared, nothing is double-counted, because two solids that meet along a plane have no common interior.

$$V_{compound} = V_1 + V_2 + \dots \text{ or } V_{solid} - V_{hole}$$

Worked case — the silo. Cylinder $r = 4$ m, $h = 12$ m, with a hemispherical cap of the same radius:

PART	FORMULA	VALUE
cylinder	$\pi r^2 h$	192π
hemisphere	$\frac{2}{3}\pi r^3$	$\frac{128\pi}{3}$
total	—	$\frac{704\pi}{3} \approx 737\text{ m}^3$

KEEP π TO THE LAST LINE

$192\pi + \frac{128\pi}{3}$ is exact; $603.2 + 134.0$ is not. Round once, at the end, and never feed a rounded number into the next part.

Which faces survive

Surface area is where the marks go. When two solids are joined, the two faces that meet **disappear** — they are now inside. The surface area of the compound is *not* the sum of the two surface areas.

$$S_{compound} = S_1 + S_2 - 2 \times (\text{shared face})$$

The silo again. The cylinder's top circle and the hemisphere's flat face are the same circle, and both are now internal:

FACE	AREA	COUNTS?
cylinder base	$\pi(16) = 16\pi$	yes — it stands on the ground
cylinder curved	$2\pi(4)(12) = 96\pi$	yes
cylinder top	16π	no — covered
hemisphere flat	16π	no — covered
hemisphere curved	$2\pi(16) = 32\pi$	yes

So $S = 16\pi + 96\pi + 32\pi = 144\pi \approx 452 \text{ m}^2$. If the silo is painted only on the outside above the ground, drop the base too: $128\pi \approx 402 \text{ m}^2$.

THREE QUESTIONS BEFORE WRITING A NUMBER

- Is the object **solid or hollow**? A bowl has an inside surface as well.
- Does it **stand on** something? Then its base may not be painted, wetted or covered.
- Which faces are **joined**? Those never count.

Bounds through a compound

Every measurement is a range. A silo measured as $r = 4 \text{ m}$ and $h = 12 \text{ m}$ to the nearest metre means $3.5 \leq r < 4.5$ and $11.5 \leq h < 12.5$.

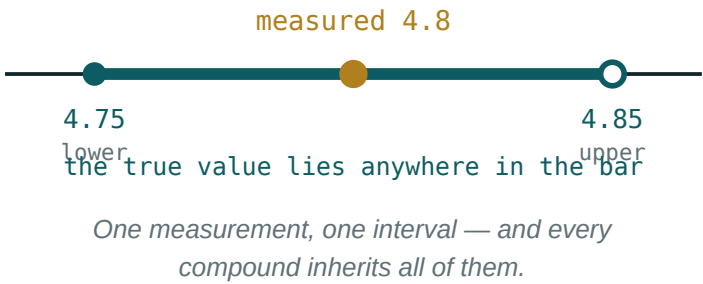
Volume grows with both, so the extremes pair up directly:

	R	H	V
lower	3.5	11.5	$\approx 532 \text{ m}^3$
stated	4	12	$\approx 737 \text{ m}^3$
upper	4.5	12.5	$\approx 986 \text{ m}^3$

A SUBTRACTION REVERSES ONE BOUND

For a pipe, $V = \pi(R^2 - r^2)L$. The **largest** metal volume needs the largest R and the **smallest** r . Whenever a quantity is subtracted, swap its two bounds.

Reporting. The silo's volume lies between 532 and 986 m³ — the two bounds do not agree even to one significant figure, so 700 m³ is the honest answer, not 737.4. Report to the accuracy the bounds support.



Measuring the classroom

Lesson 50 is practical. In pairs, choose one object in the school, measure it, and produce a single A4 sheet:

- 1. a sketch, decomposed into named solids, with every measurement marked;
- 2. the instrument used and its precision (“tape, nearest 0.5 cm”);
- 3. the volume, exact then rounded;
- 4. the surface area, saying which faces you excluded and why;
- 5. upper and lower bounds, and the accuracy you can honestly claim.

OBJECT	DECOMPOSITION	WATCH FOR
a waste bin	frustum of a cone	open top — no lid area
a football	sphere	measure the circumference, not the radius
a chalk stick	cylinder, or a frustum	the taper is real
the door frame	three cuboids	the corners are shared
a water bottle	cylinder + frustum + cylinder	capacity ≠ outer volume
a lamp shade	frustum, curved surface only	two circles missing

CAPACITY IS NOT VOLUME

A bottle's **capacity** uses inner dimensions; its **volume** as a solid object uses outer ones. The difference is the glass, and in an exam the word chosen tells you which one is wanted.

02 Worked examples

EXAMPLE 1 An ice-cream is a cone of radius 3 cm and height 10 cm topped by a hemisphere of the same radius. Find its volume.

1 Cone $\frac{1}{3}\pi(9)(10) = 30\pi$.

2 Hemisphere $\frac{2}{3}\pi(27) = 18\pi$.

3 $V = 48\pi$
Add — nothing overlaps.

4 $\approx 151 \text{ cm}^3$.

ANSWER

$48\pi \approx 151 \text{ cm}^3$

EXAMPLE 2 Find the surface area of the same ice-cream (the cone is open at the top, under the hemisphere).

1 Slant $l = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$.

2 Cone curved $\pi rl = 3\pi\sqrt{109} \approx 98.4$.
No base circle — it is joined.

3 Hemisphere curved $2\pi(9) = 18\pi \approx 56.5$.
Its flat face is joined too.

4 $S \approx 155 \text{ cm}^2$.

ANSWER

$\approx 155 \text{ cm}^2$

EXAMPLE A wooden bead is a sphere of radius 5 mm drilled through the centre by a hole of radius 2 mm.
 3 Find the volume of wood.

① Sphere $\frac{4}{3}\pi(125) = \frac{500\pi}{3}$.

② The hole is a cylinder of length 10 mm: $\pi(4)(10) = 40\pi$.
 An approximation — the true hole has spherical ends.

③ $V \approx \frac{500\pi}{3} - 40\pi = \frac{380\pi}{3}$

④ $\approx 398 \text{ mm}^3$.

ANSWER

$$\frac{380\pi}{3} \approx 398 \text{ mm}^3$$

EXAMPLE A cylindrical tank has radius 2.0 m and height 5.0 m, both to the nearest 0.1 m. Find the bounds of its capacity in litres.
 4

① $1.95 \leq r < 2.05, 4.95 \leq h < 5.05$.

② $V_{\min} = \pi(1.95^2)(4.95) \approx 59.13 \text{ m}^3$.

③ $V_{\max} = \pi(2.05^2)(5.05) \approx 66.68 \text{ m}^3$.

④ Between 59 100 and 66 700 litres.
 So “about 60 000 litres”.

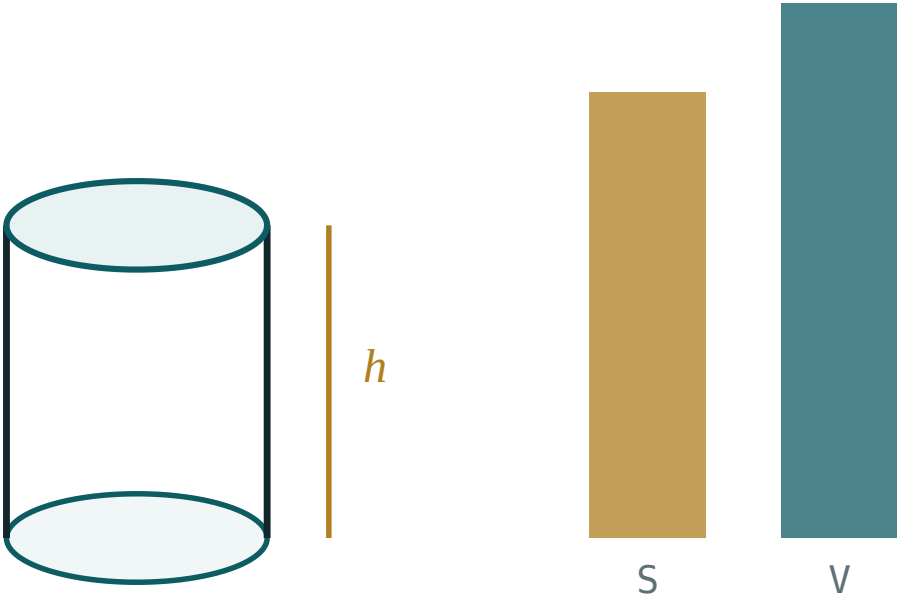
ANSWER

$\approx 59\,100$ to $66\,700$ litres

03 Interactive model

Bring a real bottle, a real ball and a tape measure. Measure in front of the class, then argue about the last digit.

A cylinder, and what a cap adds



The diagram shows a 3D cylinder with a light blue top and bottom circular face and a dark teal side. To its right is a vertical orange line segment labeled h . Further right are two vertical bars: a shorter orange bar labeled S and a taller teal bar labeled V .

base area B 50.3

perimeter P 25.1

total S 251.3

volume V 301.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Compound solid	Murakkab jism	Составное тело
Decomposition	Qismlarga ajratish	Разбиение
Hollowed solid	Bo'shliqli jism	Полое тело
Capacity	Sig'im	Вместимость
Density	Zichlik	Плотность
Upper bound	Yuqori chegara	Верхняя граница
Lower bound	Quyi chegara	Нижняя граница
Degree of accuracy	Aniqlik darajasi	Степень точности
Wetted surface	Ho'llanadigan sirt	Смачиваемая поверхность
Silo	Silos minorasi	Силосная башня

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

When two solids are joined, the shared face:

counts twice

counts once

does not count at all

is halved

Volumes of joined solids are:

added

multiplied

averaged

subtracted

For the largest metal volume in a pipe take:

both bounds large

largest R , smallest r

both small

the stated values

A silo is a cylinder plus:

a cone

a hemisphere

a prism

a frustum

Capacity uses:

outer dimensions

inner dimensions

either

the mean

Rounding should happen:

at each step

only at the end

never

twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Cylinder $r = 2$, $h = 5$ plus hemisphere $r = 2$: volume

ANSWER $20\pi + \frac{16\pi}{3} \approx 79.6$

2. Cone $r = 3$, $h = 4$ plus hemisphere $r = 3$: volume

ANSWER $12\pi + 18\pi = 30\pi$

3. Pipe $R = 5$, $r = 3$, $L = 20$: metal volume

ANSWER 320π

4. Cube edge 6 with a cylindrical hole $r = 1$ through it: volume

ANSWER $216 - 6\pi \approx 197$

5. Cylinder $r = 3$, $h = 10$ with a cone cap $h = 4$: volume

ANSWER $90\pi + 12\pi = 102\pi$

6. Bounds of 8.0 cm to the nearest 0.1 cm

ANSWER 7.95 and 8.05

7. Surface area of a solid hemisphere on a cylinder base $r = 2$, $h = 3$ (closed bottom)

ANSWER $4\pi + 12\pi + 8\pi = 24\pi$

Medium · the standard question

7 PROBLEMS

1. Silo: cylinder $r = 4$, $h = 12$ plus hemisphere. Volume

ANSWER $\frac{704\pi}{3} \approx 737 \text{ m}^3$

2. Same silo: painted outside, base excluded

ANSWER $128\pi \approx 402 \text{ m}^2$

3. Ice-cream: cone $r = 3$, $h = 10$ plus hemisphere. Surface area

ANSWER $\approx 155 \text{ cm}^2$

4. A bead: sphere $R = 5$ drilled by $r = 2$ (straight hole). Wood volume

ANSWER $\frac{380\pi}{3} \approx 398$

5. Tank $r = 2.0$, $h = 5.0$ m to 0.1 m: capacity bounds in litres

ANSWER $\approx 59\,100$ to $66\,700$

6. A dome: hemisphere $R = 6$ hollowed to $r = 5$. Material volume

ANSWER $\frac{2\pi}{3}(216 - 125) \approx 191$

7. A pencil: cylinder $r = 0.4$, $h = 15$ plus cone $h = 2$ cm. Volume

ANSWER $2.4\pi + 0.107\pi \approx 7.87 \text{ cm}^3$

Hard · stretch and reason

7 PROBLEMS

1. A silo of the same shape must hold 1000 m^3 with $r = 4$. Find h

ANSWER $h \approx 17.2 \text{ m}$

2. Pipe $R = 9$, $r = 7 \text{ cm}$, $L = 200 \text{ cm}$, all to the nearest cm: bounds of the metal volume

ANSWER $\approx 18\,200$ to $\approx 22\,200 \text{ cm}^3$

3. A cone of radius r is capped by a hemisphere of radius r . For what h are the two volumes equal?

ANSWER $h = 2r$

4. An open lamp shade is a frustum, radii 10 and 20 , slant 26 . Find the material area

ANSWER $780\pi \approx 2450 \text{ cm}^2$

5. A bottle: cylinder $r = 3$, $h = 12$, frustum $3 \rightarrow 1.5$, $h = 4$, neck $r = 1.5$, $h = 3$. Capacity in ml

ANSWER $\approx 421 \text{ ml}$

6. A hemispherical bowl of inner radius 9 cm is filled to a depth of 4 cm . Find the volume of water

ANSWER $\frac{\pi}{3}(4^2)(27 - 4) \approx 385 \text{ cm}^3$

7. A solid is a cube of edge $2R$ with the inscribed sphere removed. Find the fraction of the cube remaining

ANSWER $1 - \frac{\pi}{6} \approx 0.476$

07 Homework

Homework — 5 tasks

Task 5 is the measured object; bring the A4 sheet and the object itself if you can.

1. A grain silo is a cylinder of radius 3 m and height 10 m with a hemispherical top. Find its volume and its outside surface area, excluding the base.

2. A pipe has outer radius 8 cm, inner radius 6 cm and length 1.5 m. Find the volume of metal and its mass at 7.8 g/cm³.
3. A tank measures $r = 1.5$ m and $h = 4.0$ m, each to the nearest 0.1 m. Find the bounds of its capacity in litres, and state the answer to a sensible accuracy.
4. An ice-cream cone of radius 3.5 cm and height 12 cm carries a hemisphere of the same radius. Find the total volume, and explain which two faces you did not include in the surface area.
5. Measure one object in your home. Produce the A4 sheet: sketch, decomposition, instrument and precision, volume, surface area, bounds.